EHDS Lab Invites Applications for Summer Internships for SUPACRO Projects

Project: Machine Learning for Ground-Level $PM_{2.5}$ Estimation & Development of Smog Hazard Index (SHI)

OBJECTIVE

Develop and validate a machine-learning model that estimates ground-level $PM_{2.5}$ concentrations from satellite aerosol (AOD) and meteorological data, and use the results to build a prototype Smog Hazard Index (SHI) that classifies smog severity from Low to Extreme for the pilot area.

DATA/ INPUTS

- **Satellite:** MODIS AOD (1 km, daily); Sentinel-5P TROPOMI NO_2 and SO_2; CAMS $PM_{2.5}$/PM_{10} reanalysis.
- **Meteorology:** ERA5 — temperature, wind speed, relative humidity, planetary boundary layer height, surface pressure.
- **Ground truth (training & validation):** $PM_{2.5}$ from provincial EPA monitoring stations, low-cost sensors, and published data (optional)

(Students will be guided on how to acquire satellite and reanalysis dataset.)

METHODOLOGY

- **Weeks 1–2:** Brief literature review on AOD→$PM_{2.5}$ retrieval; assemble a matched satellite–meteorology–ground dataset; exploratory data analysis and quality checks.
- **Weeks 3–4:** Feature engineering; train and compare models — Random Forest, XGBoost and a linear baseline — using spatial–temporal cross-validation.
- **Weeks 5–6:** Evaluate models (R^2, RMSE, bias) on held-out stations; generate gridded $PM_{2.5}$ maps; build a prototype SHI by combining standardized $PM_{2.5}$, AOD, NO_2/SO_2 and meteorological stagnation indicators; classify Low → Extreme.
- **Weeks 7–8:** Validate the SHI against known historical smog episodes; produce seasonal SHI and hotspot maps; document code and write a short methodology note.

EXPECTED DELIVERABLES

- A validated $PM_{2.5}$ estimation model with a short performance report.
- Gridded $PM_{2.5}$ estimates and prototype SHI / hotspot maps for the pilot area.
- A reproducible, documented code repository (Git).
- A methodology note recommending an SHI formulation for the full four-province study (Punjab, KP, Upper Sindh, ICT).
- Preparation and submission of a journal paper.

TOOLS & SKILLS

Python, scikit-learn / XGBoost, pandas / geopandas, geospatial visualization (Matplotlib), Git.

MONITORING/ EVALUATION

- Introductory seminar (online) will be conducted to brief students on the work.
- Week-wise performance will be evaluated through online meetings.